

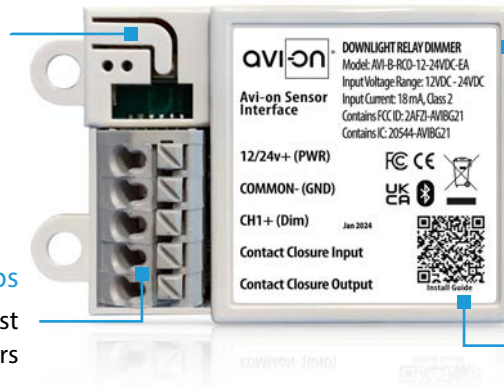


Downlight Relay Controller

Small Form Factor Dimmer for Downlights

Status LEDs & Config Button
Quick and Easy validation of wiring
and network

Terminal Blocks & Mounting Tabs
Faster, Easier and Lower Cost
Installation eliminating connectors



Dimming

Dim to off with any 1-10V Fixture
via Relay. Connect to LED string DC
output driver side to work with any
12-24V constant current driver up
to 1000mA load

External Antenna

Flexible antenna options if
mounted in metal enclosures

PRODUCT OVERVIEW

Description

The DC Downlight Relay Dimmer provides a 1000 mA DC output contact closure relay, 0-10V dimming output, and Direct Connect® Sensor port. This flexible module can be used as a small form factor controller for downlights and other small fixtures that do not have dim to off drivers. The controller can also be used as general purpose contact closure output without the dimming lines to drive output commands to external relays, building control systems, or any other use requiring contact closure output.

Relay Dimmers are part of the Avi-on™ Bluetooth with Mesh product ecosystem supported by the Avi-on mobile app, commissioning tools, and cloud IoT Services.

Any output command from the Avi-on network can be directed to the Relay Dimmer including motion sensors, daylight harvesting, wall stations, and schedules

The Downlight Relay Dimmer comes with an external antenna for flexible mounting options in metal enclosures.

Applications

DC Fixture Controller

The Downlight Relay Controller works with 12~24V DC, controlling fixtures up to 1A. The small form factor allows it to fit within the wiring box of individual downlights, and can also support a Direct Connect sensor to add motion sensing and daylight control for downlights. The relay can disconnect power to the LEDs to support drivers that do not dim to off. The relay can be wired on the LED (DC side of the driver) to support AC constant current 1-10v drivers with less than 24V max voltage and 1A maximum current. See wiring diagram below.

Contact Closure Output

The Downlight Relay Controller can also be used as a generic contact closure output for control of circuit level relays in control panels (DC powered), external commands to building control systems, VAV boxes, custom indicator lights, or other uses that need an on/off signal based on a command from the Avi-on Network.

ORDERING INFORMATION

Part Number	Description	Application	Input Voltage
AVI-B-RCO-12-24VDC-EA	Downlight Relay Dimmer	Indoor	12 - 24 VDC

To order please contact Avi-on sales at (877) AVION-US, (877) 284-6687 or prosales@avi-on.com
for order information or integration support.

Project	Location/ Type
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SPECIFICATIONS

Input Voltage:	12VDC - 24VDC
Current Draw:	25mA @12V, 30mA @24V
Relay Capacity:	30 VDC @ 1A
Switching Current:	2A Max
Mounting:	Removable mounting tabs
Weight:	0.45 oz (16g)
Terminal Blocks:	22-16 AWG wires
Operating Temp*:	-40C to +70C (-40F to +158F)
Storage Temperature:	-40C to +85C (-40F to +185F)
Humidity Rating:	95% non-condensing
Radio Frequency:	2.4GHz
Wireless Standard:	BLE 4.2 with Mesh

Note: Operating controller at or above these levels may result in damage to components/product

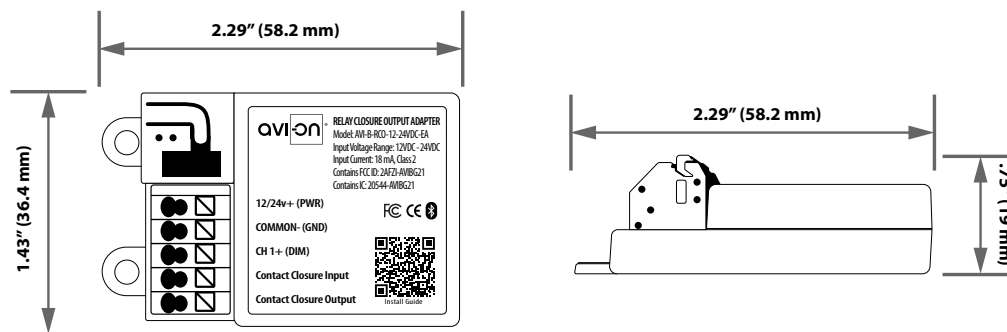
Point to Point Range**	80ft with obstructions; 350ft unobstructed
ESD:	IEC/EN 61000-4
Security:	AES 128-bit encryption for device to device communication AES 256-bit encryption for device to cloud communication
Warranty:	5 years; 10 years optional
Certifications:	FCC ID: 2AFZI-AVIBG21 IC ID: 20544-AVIBG21 BQB: DID: D063032 CE: EN IEC 6100-6-1:2019, EN IEC 6100-6-3:2021 UL 916, 2043 (Plenum Rated) UL 924 Emergency Lighting Equipment

*For temperatures below freezing or dew point, the device should be sealed in a watertight container to prevent frost buildup

**When communicating through the mesh, range is essentially unlimited (5000ft+)

See Safety and Certification Details here: <https://tinyurl.com/y4698945>

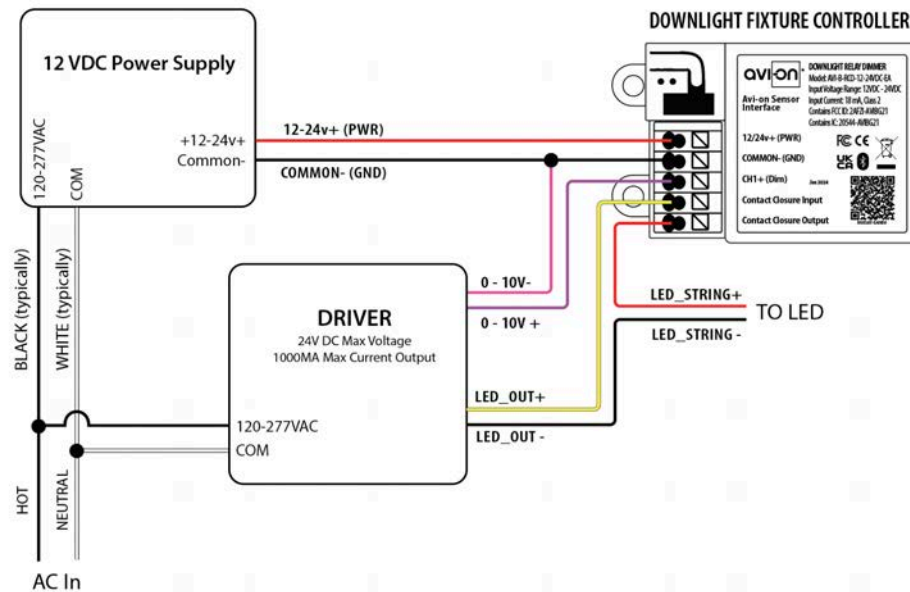
DIMENSIONS



WIRING DIAGRAMS

Downlight Fixture Controller Wiring

Use DC Power supply to provide 12-24VDC power to the Controller. Wire the 0-10V CH1+ line to driver dimming lines. Connect the LED Driver OUTPUT (not AC Input) positive wire to the Contact Closure Input terminal, and the Contact Closure Output Terminal to the LED string Positive line. Plug a Direct Connect Sensor in the sensor port if used. **Driver must not exceed 24V maximum voltage or 1A (1000MA) current.**



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